

## Café Sales Data Analysis Project using SQL and Power BI

About the dataset:

The café sales dataset is from Kaggle. It is a synthetic dataset representing sales transactions in a café. This dataset is intentionally dirty to provide a realistic scenario for Data Cleaning and Exploratory Data Analysis (EDA). This dataset will serve as my practice and a project for my portfolio.

Link: <https://www.kaggle.com/datasets/ahmedmohamed2003/cafe-sales-dirty-data-for-cleaning-training>

The steps I have taken for this project are Data Cleaning using SQL, which includes fixing the dataset for it to be ready for Exploratory Data Analysis (EDA). Then I performed EDA to find patterns and insights hidden in the dataset using SQL also. Lastly, the last step I did is to create a dashboard out of my cleaned dataset to visualize and to prepare the data for reporting using Power BI.

| Transaction ID | Item     | Quantity | Price Per Unit | Total Spent | Payment Method | Location | Transaction Date |
|----------------|----------|----------|----------------|-------------|----------------|----------|------------------|
| TXN_1961373    | Coffee   | 2        | 2              | 4           | Credit Card    | Takeaway | 9/8/2023         |
| TXN_4977031    | Cake     | 4        | 3              | 12          | Cash           | In-store | 5/16/2023        |
| TXN_4271903    | Cookie   | 4        | 1              | ERROR       | Credit Card    | In-store | 7/19/2023        |
| TXN_7034554    | Salad    | 2        | 5              | 10          | UNKNOWN        | UNKNOWN  | 4/27/2023        |
| TXN_3160411    | Coffee   | 2        | 2              | 4           | Digital Wallet | In-store | 6/11/2023        |
| TXN_2602893    | Smoothie | 5        | 4              | 20          | Credit Card    |          | 3/31/2023        |
| TXN_4433211    | UNKNOWN  | 3        | 3              | 9           | ERROR          | Takeaway | 10/6/2023        |
| TXN_6699534    | Sandwich | 4        | 4              | 16          | Cash           | UNKNOWN  | 10/28/2023       |
| TXN_4717867    |          | 5        | 3              | 15          |                | Takeaway | 7/28/2023        |
| TXN_2064365    | Sandwich | 5        | 4              | 20          |                | In-store | 12/31/2023       |
| TXN_2548360    | Salad    | 5        | 5              | 25          | Cash           | Takeaway | 11/7/2023        |
| TXN_3051279    | Sandwich | 2        | 4              | 8           | Credit Card    | Takeaway | ERROR            |

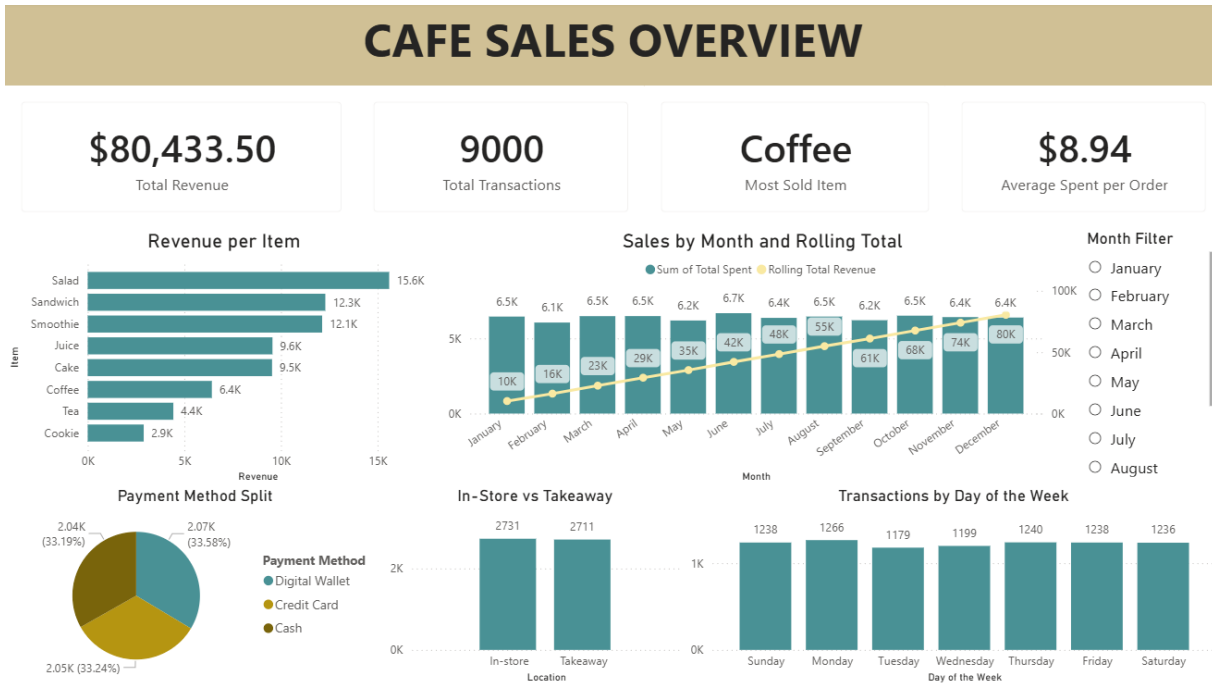
**Figure 1. Snapshot of the raw dataset**

Looking at the table, I can already see some missing values, unknowns and errors for each column except the Transaction ID. There are no duplicate records in this dataset so I started cleaning by filling the blanks with unknowns for non computable columns like item, location and payment method. For total spent I multiplied quantity to price per unit to get the value. For missing values in the item column, I couldn't fill it using the price per unit because there are items that are same in price, so I left it as unknown. For the dates I changed the error/unknown values to null. Lastly, I removed rows with little to no data, for example, rows with item, payment method, location and transaction date columns are blank or error/unknown. I completely removed those rows since it provides no value.

| Transaction ID | Item     | Quantity | Price Per Unit | Total Spent | Payment Method | Location | Transaction Date |
|----------------|----------|----------|----------------|-------------|----------------|----------|------------------|
| TXN_1961373    | Coffee   | 2        | 2              | 4           | Credit Card    | Takeaway | 9/8/2023         |
| TXN_4977031    | Cake     | 4        | 3              | 12          | Cash           | In-store | 5/16/2023        |
| TXN_4271903    | Cookie   | 4        | 1              | 4           | Credit Card    | In-store | 7/19/2023        |
| TXN_7034554    | Salad    | 2        | 5              | 10          | UNKNOWN        | UNKNOWN  | 4/27/2023        |
| TXN_3160411    | Coffee   | 2        | 2              | 4           | Digital Wallet | In-store | 6/11/2023        |
| TXN_2602893    | Smoothie | 5        | 4              | 20          | Credit Card    | UNKNOWN  | 3/31/2023        |
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| TXN_4717867    | UNKNOWN  | 5        | 3              | 15          | UNKNOWN        | Takeaway | 7/28/2023        |
| TXN_2064365    | Sandwich | 5        | 4              | 20          | UNKNOWN        | In-store | 12/31/2023       |
| TXN_2548360    | Salad    | 5        | 5              | 25          | Cash           | Takeaway | 11/7/2023        |
| TXN_3051279    | Sandwich | 2        | 4              | 8           | Credit Card    | Takeaway | NULL             |

**Figure 2. Cleaned Dataset Snapshot**

Note: I took the snapshot of the csv files but I did the cleaning in SQL (check file).



**Figure 3. Café Sales Dashboard**

For the EDA, I tried to answer common business questions like the timeline of the dataset which is the whole year of 2023, total revenue of \$80,433.50 which is the total money earned by the cafe, total orders/transactions which is 9,000 tells us the number of orders processed, best seller/most sold item which is coffee as expected in a coffee shop, average spent per order tells me that customers spend \$9 for each order, each item ranges from \$1 to \$5, this average means that customers are buying more than 1 item per order, meaning that we could push prices to reach a

goal like \$10 average per order, we could also determine if customers are buying less if this number drops. Although coffee is the most sold item, the salad has the highest revenue produced because of its price of \$5 each so it's possible that customers are pairing salad with other items on the menu to get an average of \$8.94. Revenue per Item, as I said salad produces the most revenue because of its high price, sandwich and smoothie followed with its price of \$4 each, cookie being the lowest priced item with \$1 each has also the lowest produced revenue. Sales by month and the rolling total shows that each month has at least \$6K in sales while the rolling total we can see that it accumulates steadily each month reaching the \$80K at the end of the year without losses or flat months which is good for a business. As for the payment method split, this dataset shows that each payment method is almost equal to each other meaning that they are being utilized correctly and customers can pay using the method they want. In-store vs takeaway locations, in this dataset, the blanks or unknowns/errors counts more than dine in or take out customers, I removed it from the visualization but it has a value of 3234 rows while dine in and take out contains at least 2.7K. There might be a problem or a bug in the Point of Sale (POS) system of the café since there are more errors/unknowns and blanks recorded than the designated values. Lastly, I have Transactions by Day of the Week where it shows the record of transactions happening per day, each month gives different patterns of customer transactions, for example, the month of May has 115 transactions on Monday and Wednesday which may be people starting the week strong or just want something to treat themselves, while on the month of December, Saturdays has the most transactions with 127 orders followed by Sunday with 120 orders which may be people taking their precious time off during the holidays or taking out their families during weekends, meaning sales may vary depending on the season.

Note: I have done the EDA in SQL and Dashboard in Power BI (check files)

To conclude this Data Analysis project, I can say that the Café business is going on the right track based on the data. Steady growth, high average spent per orders and consistent sales for the whole year. There might be an error on the Point of Sale system (POS) because of the number of transactions where unknown/error or blank is higher than the values set for the location. I also want to take note that this is a synthetic dataset and this is how I would analyze when given a real data also. I made use of tools like SQL, Power Query on Power BI and Power BI to complete this project.